

Where DGLAP comes from

It is not about how many gluons you measure — it is about which ones you don't

$$\zeta = k^+/S \text{ the measured daughter's momentum fraction of the parent } S = p^+ + k^+; C_{UV}(\zeta) = \frac{\zeta}{\zeta} + \frac{\bar{\zeta}}{\zeta} + \zeta\bar{\zeta} = P_{gg}(\zeta)/2N_c$$

No — and the logic runs the other way. Collinear (DGLAP) logarithms are produced by **unresolved** partons. Measuring a parton fixes its momentum and thereby *removes* it from the set that can generate a collinear log. So integrating one final-state gluon out is not what *prevents* fragmentation — it is the very thing that *enables* it. **Your own Row 8 proves it:** single-inclusive gluon production, one gluon measured, one integrated out, and it produces the complete $P_{gg}(\zeta)$ (§1). The reason N_2 does not is specific to N_2 's topology, not to single-inclusive kinematics (§2).

1. The counterexample you already computed: Row 8

Row 8 is the row with two $\mathfrak{C}$ vertices. Its kinematics are exactly the ones you are asking about: $\mathfrak{A}^\dagger$ creates **one** gluon of momentum $S = p^+ + k^+$ from ρ , and $\mathfrak{C}$ splits it into the **measured** gluon (k^+) and the **unobserved** one (p^+), which is integrated out. Doing the z integral in $d = 2 - 2\epsilon$ and assembling:

$$\frac{\pi}{k^{+2}} \left\{ C_{UV}(\zeta) \mathcal{L} \mathcal{X} + \zeta \bar{\zeta} [\mathcal{X} - 2\mathcal{R}] \right\}$$

with $\mathcal{L} = \frac{1}{\epsilon} - \log(4\pi^2 \mu^2 r^2)$ the $\overline{\text{MS}}$ transverse log, and the symbolic difference against the direct expansion exactly 0. The coefficient of the transverse log is the **full** splitting function — all three pieces $\frac{\zeta}{\zeta} + \frac{\bar{\zeta}}{\zeta} + \zeta\bar{\zeta}$, including the regular $\zeta\bar{\zeta}$. The $\frac{2}{k^+}$ term dies by $\zeta + \bar{\zeta} = 1$, which is the momentum sum rule. That is a DGLAP log, in single-gluon production, with one final-state gluon integrated out.

And its kinematics are the fragmentation ones: the $\mathfrak{C}$ delta ties the emission points as $w' - w = \frac{S}{k^+}(y' - y) = \frac{1}{\zeta}(y' - y)$, so the phase is $e^{-i(k_\perp/\zeta) \cdot (y' - y)}$ — the hard factor sits at the **parent's** transverse momentum $k_\perp/\zeta$, with measure $\frac{d\zeta}{\zeta^2}$.

2. So what is actually different about N_2 ?

Not the number of measured gluons — both rows measure one and integrate one out. The difference is whether the amplitude's and the conjugate's splitting kernels become singular on the **same** transverse separation:

	Row 8 (two $\mathfrak{C}$)	N_2 (the $ \mathfrak{B}_2 ^2$ piece)
emission points	slaved: $w' - w = (y' - y)/\zeta$	free: w and w' integrated independently
phase	$e^{-i(k_\perp/\zeta)(y' - y)}$ — parent's $k_\perp$	$e^{-ik_\perp(w' - w)}$ — observed $k_\perp$, unrescaled
collinear region	both kernels singular on $y' - y$: $\rho^{-2} \Rightarrow \log$	singular on $z - w$ and $z - w'$, different: $\rho^{-1} \Rightarrow$ no log
ζ weight of the log	$C_{UV}(\zeta) = P_{gg}/2N_c$, complete	$1/(\zeta\bar{\zeta})$, soft ends only
Wilson lines see	the parent (split after the shockwave)	one daughter per term: $U(w)$ or $U(z)$
what the log is	collinear $\rightarrow$ fragmentation function	rapidity $\rightarrow$ JIMWLK / BK

The last row is the physical statement. A fragmentation log needs the target to scatter off the **parent**, with the splitting happening afterwards — then the observed hadron carries a fraction ζ of what the target actually saw, and the unresolved sibling's transverse momentum sweeps the collinear region. In N_2 neither term has a Wilson line at a common parent position: term 1 has $U^{ac}(w)$ on the observed daughter, term 2 has $U^{bc}(z)$ on the unobserved one. That is the eikonal, one-on-each-side-of-the-shockwave configuration, and it produces a rapidity log — which is exactly what came out.

3. Why measuring both gluons would make it worse, not better

A collinear logarithm is $\int \frac{d^2 q_\perp}{q_\perp^2}$ over the relative transverse momentum of a splitting. That integral only exists if $q_\perp$ is **integrated**. If you measure both gluons, the pair's relative momentum is fixed by the measurement: there is no $q_\perp$ integral left, the splitting is fully resolved, and no collinear log arises from that pair at all. What you get instead is a finite two-particle correlation with a steep (but integrable in the measured variables) enhancement at small opening angle.

At NLO for two–gluon production the collinear logs come back — but from the **third**, unobserved parton, not from the measured pair. The rule is completely general:

Collinear logs are generated by unresolved partons only. Each measurement you impose removes one parton from that set. Single–inclusive production at NLO has exactly one unresolved parton, and it is precisely that parton which generates the three collinear/rapidity structures the calculation must contain.

4. The three logs single–inclusive production has to produce

For the NLO cross section to factorise at all, all three must be there — in single–gluon production, with one gluon integrated out:

log	absorbed into	signature	seen in
$\log \mu^2$, coefficient $P_{gg}(\zeta)$, hard factor at $k_\perp/\zeta$, measure $\frac{d\zeta}{\zeta^2}$	fragmentation function $D(\zeta, \mu)$	$\mathcal{C}$ ties $w' - w = (y' - y)/\zeta$	Row 8
$\log \mu^2$, coefficient P_{gg} , measure $\frac{d\zeta}{\zeta}$, kernels at the source	projectile distribution	splitting before the shockwave, both daughters' lines	to be identified
$\log \frac{1}{\Lambda}$, coefficient $\propto \frac{1}{\zeta\bar{\zeta}}$ (soft only), LO transverse structure	JIMWLK / BK evolution of the target	z –gluon decouples; $\int \frac{d^2z}{ z ^2}$	N_2 , and the other real rows

So N_2 is not a row that "failed" to give DGLAP — it is a row that belongs to the third line. Different rows carry different logs; no single row has to carry all of them.

5. The practical test

You do not have to guess which row is which. Two diagnostics settle it, and both are cheap:

- Power counting on the candidate collinear separation.** A log needs the integrand $\sim \rho^{-2}$ there, i.e. both the amplitude and the conjugate kernel singular on the *same* variable. ρ^{-1} means no log.
- The ζ weight.** Strip the transverse factor and vary ζ at fixed S . If the coefficient tracks $C_{UV}(\zeta) = \zeta\bar{\zeta} + \frac{1}{\zeta\bar{\zeta}} - 2$ it is DGLAP; if it is $\frac{1}{\zeta\bar{\zeta}}$ alone (or $\frac{1}{\zeta}$ alone) it is soft, hence a rapidity log. The missing $\zeta\bar{\zeta} - 2$ is the tell — the regular part of P_{gg} is what no soft approximation can produce.

Row 8 passes both. N_2 fails both, in exactly the way a soft region should.